1. Introduction

The rapid advancement of generative AI has created an unp
Publishers who invest in original journalism find their work indi
Regulatory frameworks like the EU AI Act and California SB 9
AI-generated content, yet enforcement remains elusive without

Content provenance offers a fundamentally different approa
attempting to classify content after the fact, provenance syster
creation. The Coalition for Content Provenance and Authentic
this purpose, originally designed for images and video. Extend
challenges: text is routinely copied, reformatted, and excerpte

In this paper, we present three key contributions. First, we d
adapted for text documents using invisible Unicode encoding.
manifest modes that dramatically reduce the overhead of prov
character steganography for persentence integrity verification,
documents are partially quoted or excerpted.

2. Background and Related Work

The C2PA specification defines a manifest structure contain
signatures that establish the provenance of digital assets. Orig
standard uses CBOR-encoded data structures signed with X.5
embedded in metadata containers such as EXIF or XMP. Text
necessitating alternative embedding strategies.

Prior work on text watermarking has explored statistical app
syntactic transformations. These methods alter the content its
Unicode steganography, by contrast, embeds information in in
text. Previous Unicode steganography schemes used variatior
cryptographic integrity guarantees and standardized verificatio

Our approach builds on these foundations by combining C2
steganography, creating a system where provenance metadat
This allows publishers to sign content at creation time while pr

3. Methodology

3.1 C2PA Manifests for Text

We encode C2PA manifests as CBOR binary data, then rep
Unicode characters. The full manifest includes the claim gene
an Ed25519 digital signature. This data is appended to the do
from the Unicode Supplementary Special-purpose Plane, spec

The full manifest approach provides maximum provenance c
overhead. For a typical 500-word article, the full C2PA manifes
representing a 40-60% increase in document size. While these
storage requirements and may be stripped by certain text proc

3.2 Lightweight and Minimal Manifests

To address the overhead of full manifests, we introduce two

manifest replaces the full CBOR payload with a 128-bit UUID p
serverside, and the UUID serves as a lookup key for verificatio
approximately 1,500 invisible characters: a 90% reduction com

The minimal manifest mode extends this concept to persegm
own UUID pointer, while a single document-level C2PA wrappe
This enables fine-grained attribution: if a single sentence is cop
within that sentence can be used to trace it back to its origin, e

3.3 Zero-Width Steganographic Embedding

For environments requiring maximum compatibility, we deve
using four carefully selected Unicode characters: Zero-Width N
Combining Grapheme Joiner (U+034F), and Mongolian Vowel
base-4 encoding alphabet, allowing us to represent arbitrary bi

Each sentence receives a 128-character signature consisting
HMAC-SHA256 truncated digest. The HMAC is computed over
signing key, binding the sentence content to its provenance re
covers the concatenation of all sentence-level signatures, crea
a Merkle tree.

This approach offers several advantages over variation sele
render invisibly across all major platforms including Microsoft W
viewers. Our testing confirmed that variation selectors, while o
boxes in Microsoft Word, making them unsuitable for enterpris

4. Results

We evaluated our embedding approaches across four dimen
verification accuracy, and robustness to common transformati
news articles ranging from 200 to 3,000 words.

The full C2PA manifest mode achieved 100% verification ac
Lightweight UUID mode reduced this overhead to 4.2% while
documents where the serverside manifest remained accessibl
overhead but enabled individual sentence verification. Zero-wid
overhead per sentence with full platform compatibility.

Robustness testing demonstrated that all embedding modes
Firefox, Safari, Microsoft Word, and Google Docs. PDF export
characters. Content management systems including WordPre
configured to allow Unicode pass-through. The primary failure
non-printable characters, which affected 12% of tested CMS co

5. Conclusion

Cryptographic content provenance provides a robust, standa
detection. By embedding invisible signatures at creation time,
authorship that persists through the content distribution lifecyc
modes make this practical for real-world deployment, with over
pipelines.

Future work will focus on three areas: extending the approac
HTML and PDF, developing browser extensions for automatic

the broader C2PA ecosystem for images and video. As regula
increase, cryptographic provenance will become an essential
ecosystem.

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